

# Topological Memory - Dependence

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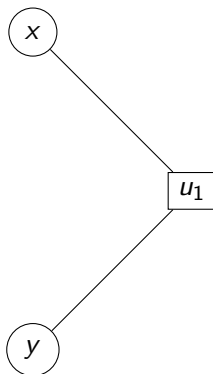
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# Introduction

Today:

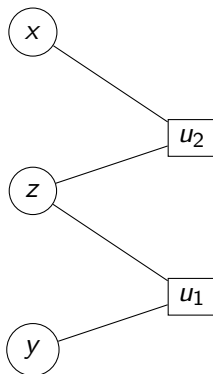
- ▶ Dependence.
- ▶ Local Stochastic Noise.

## Dependence Type I



$$\begin{aligned} & \mathbf{Pr}[(x, y) \in e | u_1 \text{ is satisfied}] \\ &= \mathbf{Pr}[x \in e | u_1 \text{ is satisfied}] \mathbf{Pr}[y \in e | u_1 \text{ is satisfied}] \\ &\leq \mathbf{Pr}[x \in e] \mathbf{Pr}[y \in e] \end{aligned}$$

## Dependence Type II



$$\begin{aligned} & \mathbf{Pr}[(x, y) \in e | z \text{ is not flipped}] \\ &= \mathbf{Pr}[x \in e | z \text{ is not flipped}] \mathbf{Pr}[y \in e | z \text{ is not flipped}] \\ &\leq \mathbf{Pr}[x \in e] \mathbf{Pr}[y \in e] \end{aligned}$$

## Definition

Consider two (qu)bits  $x, y$ . We say that  $z$  is a **linked node** between  $x$  and  $y$  if  $z$  is either a bit or a check that is shared by both of their  $D$  light cones.

## Remark

Notice that  $y$  might be a **linked node** between  $x$  and  $y$ .

## Definition

Consider an application of a noise channel. We say that  $z$  is a **clean linked node** if:

1.  $z$  is a bit and  $z \notin f$  (not flipped).
2.  $z$  is a check and  $z$  is satisfied.

$$\begin{aligned}
& \Pr[x_1, x_2, \dots, x_k, x_{k+1}, \dots, x_l \in e \mid z_1, z_2, \dots, z_m] \\
&= \prod_i \Pr[x_i \in e \mid z_1, z_2, \dots, z_m] \\
&= \prod_i \Pr[x_i \in e \mid z \in \Gamma(x_i)] \\
&\leq \prod_{i=1}^k \Pr[x_i \in e] \prod_{i=k+1}^l \Pr[x_i \in e \mid z_i \in \Gamma(x_i)] \\
&\leq M(p)^k
\end{aligned}$$

# McDiarmid's Inequality - Local Stochastic Noise

## $C_1, C_2$ - local stochastic noise

Let  $C_1, C_2 > 0$ . We say that  $\rho$  is subjected to  $C_1, C_2$ - local stochastic noise, if for any subset of qubits  $S$  the probability they will be faulty is bounded by:

$$C_1 p^{|S|} \leq \Pr[S \in e] \leq C_2 p^{|S|}$$

## Lemma: McDiarmid's Inequality

McDiarmid's Inequality is correct for  $C_1, C_2$ - local stochastic noise.

# Proof

## Proof

$$\star = \sup_{x, \bar{x}} \mathbf{E} [f | X_1, \dots, X_{i-1}, x] - \mathbf{E} [f | X_1, \dots, X_{i-1}, \bar{x}]$$

First we show that:

$$\begin{aligned} \frac{C_2}{C_1} p^{|Z|} &\leq \Pr [z_{i+1}, \dots, z_n | X_1, \dots, X_{i-1}, x] \leq \frac{C_2}{C_1} p^{|Z|} \\ \frac{C_2}{C_1} p^{|Z|} &\leq \Pr [z_{i+1}, \dots, z_n | X_1, \dots, X_{i-1}, \bar{x}] \leq \frac{C_2}{C_1} p^{|Z|} \end{aligned}$$

And then we get that:

$$\star \leq \sum_Z p^{|Z|} \left( \frac{C_2}{C_1} - \frac{C_1}{C_2} \right) c$$



# Local Decoder

## definition

We say that  $D$  is a local decoder if when applied against  $2p(1-p)$ -depolarized channel:

- ▶ There exists function  $M : (0, 1) \rightarrow (0, 1)$  such that the probability of each bit being flipped is  $M(2p(1-p)) < p^{\frac{100}{99}}$ .
- ▶ There is a constant  $l$  such that for any bit, there are only  $(\Delta)^l$  bits on which the event of it being flipped depends.

For example, if we take  $k$  large enough and consider the  $(n, k)$ -Toric, then both the swift-rule and the flip upon majority are local decoders.

## Definition

Consider the moment after the application of the depolarizing channel, and right before the correction by the decoder.

For a subset of qubits  $S$ , we say that a qubit  $x \in S$  is depended if there exists  $y \in S$  such:

1. Either  $x$  and  $y$  share a check that is not satisfied. (Type I)
2. Or,  $x$  and  $y$  have checks which share a bit that was flipped by the application of the channel. (Type II)